

### (a) Controlled Instrument

Frozen Recipe: SGD 0.1, Cosine,  
200 Epochs, Batch 128, Crop and Flip

Committed Subset Indices,  
Per Dataset and Fraction

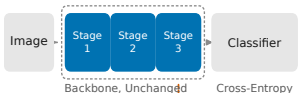
Pinned Filter Bank (Fingerprinted)

Every Intervention Sees

**Byte-Identical Images**

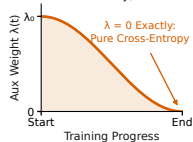
Deviations Quarantined, Never Headline

### (b) MomentAux: The Fused Prior

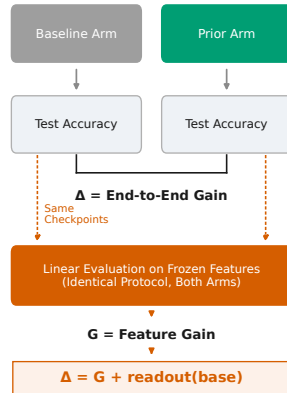


+2% Training Compute, +0 at Inference

Prior Dominates Early, Then Vanishes



### (c) Two Measurements Per Cell



### (d) Comparators, By Cost

ImageNet Transfer  
External Data

SimCLR, SimSiam, DINO  
2× Compute

FitNets Learned Teacher  
2× Compute

DeiT Augmentation  
Recipe Change

HOG Target  
1.02×

MomentAux (Ours)  
1.02×

All Fused Into the Same Frozen Recipe;  
Combinations Measured Too